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Source/drain regions of semiconductor devices and methods of forming the same

Abstract

A semiconductor device includes: a first fin and a second fin extending from a substrate and an epitaxial source/drain region. The epitaxial source/drain region includes a first portion grown on the first fin and a second portion grown on the second fin, and the first portion and the second portion are joined at a merging boundary. The epitaxial source/drain region further includes a first subregion extending from a location level with a highest point of the epitaxial source/drain region to a location level with a highest point of the merging boundary, a second subregion extending from the location level with the highest point of the merging boundary to a location level with a lowest point of the merging boundary, and a third subregion extending from the location level with the lowest point of the merging boundary to a location level with a top surface of an STI region.

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Background/Summary

PRIORITY CLAIM AND CROSS-REFERENCE (1) This application is a continuation of U.S. patent application Ser. No. 17/712,897, filed on Apr. 4, 2022 and entitled, “Source/Drain Regions of Semiconductor Devices and Methods of Forming the Same,” which is a divisional of U.S. patent application Ser. No. 16/901,791, filed on Jun. 15, 2020, now U.S. Pat. No. 11,296,080 issued Apr. 5, 2022, and entitled, “Source/Drain Regions of Semiconductor Devices and Methods of Forming the Same,” each application is hereby incorporated herein by reference.

BACKGROUND

(1) Semiconductor devices are used in a variety of electronic applications, such as, for example, personal computers, cell phones, digital cameras, and other electronic equipment. Semiconductor devices are typically fabricated by sequentially depositing insulating or dielectric layers, conductive layers, and semiconductor layers of material over a semiconductor substrate, and patterning the various material layers using lithography to form circuit components and elements thereon.

(2) As the semiconductor industry has progressed into nanometer technology process nodes in pursuit of higher device density, higher performance, low power consumption, and lower costs, challenges from both fabrication and design issues have resulted in the development of three-dimensional designs, such as a Fin Field Effect Transistor (FinFET). FinFET devices typically include semiconductor fins in which channel and source/drain regions are formed. A gate is formed over and along the sides of the fin structure (e.g., wrapping) utilizing the advantage of the increased surface area of the channel to produce faster, more reliable, and better-controlled semiconductor transistor devices. However, with the decrease in scaling, new challenges are presented to IC fabrication.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

(2) FIG. 1 illustrates an example of a FinFET in a three-dimensional view, in accordance with some embodiments.

(3) FIGS. 2, 3, 4, 5, 6, 7, 8A, 8B, 9A, 9B, 10A, 10B, 10C, 10D, 10E, 11A, 11B, 11C, 11D, 12A, 12B, 12C, 12D, 13A, 13B, 13C, 13D, 14A, 14B, 14C, 14D, 14E, 15A, 15B, 15C, 15D, 16A, 16B, 16C, 16D, and 16E are cross-sectional views of intermediate stages in the manufacturing of FinFETs, in accordance with some embodiments.

DETAILED DESCRIPTION

(4) The following disclosure provides many different embodiments, or examples, for implementing different features of the invention. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in

itself dictate a relationship between the various embodiments and/or configurations discussed.

(5) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(6) Aspects of the present disclosure relate to an epitaxial scheme for a source/drain region in a semiconductor device, such as an n-type Field Effect Transistor (nFET), which may be a Fin Field Effect Transistor (FinFET) device. Source/drain regions of transistors, for example, and methods for forming such features are described. Techniques and apparatus are provided herein for forming source/drain regions in a semiconductor device having a rounded top profile. The techniques may reduce nodule defects by choice of carrier gas and an optimized gas ratio. In embodiments in which the FinFETs are used in memory arrays including high current (HC) static random access memory (SRAM) areas, the rounded top shapes of the epitaxial source/drain regions may prevent non-merging of intra-fin epitaxial regions, e.g. silicon phosphide (SiP), of neighboring source/drain regions. In embodiments in which the FinFETs are used in memory arrays including high density (HD) SRAM areas, the rounded top shapes of the epitaxial source/drain regions may improve fin coverage of HD SRAM structures and prevent merging, or shorts, between neighboring HD SRAM source/drain regions by creating slimmer epitaxial source/drain shapes. The rounded top profiles of the epitaxial regions can permit a larger landing area for a contact for both HC and HD SRAM structures, which may further reduce contact resistance. The rounded top profiles may reduce highly doped SiP source/drain region loss for downstream middle end of line (MEOL) and back end of line (BEOL) processes. Total production throughput may improve by about 20% due to a higher epitaxial growth rate and a reduced transition time. The higher intra-fin merge height of the HC SRAM structures and the slim epitaxial source/drain region shapes of the HD SRAM structures may improve device performance by reducing the capacitance effect of the source/drain regions. Enlarged highly doped source/drain volume may lead to reduced source/drain contact plug resistance.

(7) Example techniques for forming the source/drain regions are described and illustrated herein with respect to Fin Field-Effect Transistors (FinFETs); however, an epitaxy scheme within the scope of this disclosure can also be implemented in other semiconductor devices. Further, intermediate stages of forming FinFETs are illustrated. Some aspects described herein are described in the context of FinFETs formed using a replacement gate process. In other examples, a gate-first process is used, as a person of ordinary skill in the art will readily understand. Some variations of the example methods and structures are described. A person having ordinary skill in the art will readily understand other modifications that may be made that are contemplated within the scope of other embodiments. Although method embodiments may be described in a particular order, various other method embodiments may be performed in any logical order and may include fewer or more steps described herein.

(8) FIG. 1 illustrates an example of a FinFET in a three-dimensional view, in accordance with some embodiments. The FinFET comprises a fin 52 on a substrate 50 (e.g., a semiconductor substrate). Isolation regions 56 are disposed in the substrate 50, and the fin 52 protrudes above and from between neighboring isolation regions 56. Although the isolation regions 56 are described/illustrated as being separate from the substrate 50, as used herein the term “substrate” may be used to refer to just the semiconductor substrate or a semiconductor substrate inclusive of isolation regions. Additionally, although the fin 52 is illustrated as a single, continuous material as the substrate 50, the fin 52 and/or the substrate 50 may comprise a single material or a plurality of materials. In this context, the fin 52 refers to the portion extending between the neighboring

isolation regions **56**.

(9) A gate dielectric layer **92** is along sidewalls and over a top surface of the fin **52**, and a gate electrode **94** is over the gate dielectric layer **92**. Source/drain regions **82** are disposed in opposite sides of the fin **52** with respect to the gate dielectric layer **92** and gate electrode **94**. FIG. **1** further illustrates reference cross-sections that are used in later figures. Cross-section A-A is along a longitudinal axis of the gate electrode **94** and in a direction, for example, perpendicular to the direction of current flow between the source/drain regions **82** of the FinFET. Cross-section B-B is perpendicular to cross-section A-A and is along a longitudinal axis of the fin **52** and in a direction of, for example, a current flow between the source/drain regions **82** of the FinFET. Cross-section C-C is parallel to cross-section A-A and extends through a source/drain region of the FinFET. Subsequent figures refer to these reference cross-sections for clarity.

(10) Some embodiments discussed herein are discussed in the context of FinFETs formed using a gate-last process. In other embodiments, a gate-first process may be used. Also, some embodiments contemplate aspects used in planar devices, such as planar FETs.

(11) FIGS. **2** through **16E** are cross-sectional views of intermediate stages in the manufacturing of FinFETs, in accordance with some embodiments. FIGS. **2** through **7** illustrate reference cross-section A-A illustrated in FIG. **1**, except for multiple fins/FinFETs. FIGS. **8A**, **9A**, **10A**, **11A**, **12A**, **13A**, **14A**, **15A**, and **16A** are illustrated along reference cross-section A-A illustrated in FIG. **1**, and FIGS. **8B**, **9B**, **10B**, **10E**, **11B**, **12B**, **13B**, **14B**, **14E**, **15B**, **16B**, and **16E** are illustrated along a similar cross-section B-B illustrated in FIG. **1**, except for multiple fins/FinFETs. FIGS. **10C**, **10D**, **11C**, **11D**, **12C**, **12D**, **13C**, **13D**, **14C**, **14D**, **15C**, **15D**, **16C**, and **16D** are illustrated along reference cross-section C-C illustrated in FIG. **1**, except for multiple fins/FinFETs.

(12) In FIG. **2**, a substrate **50** is provided. The substrate **50** may be a semiconductor substrate, such as a bulk semiconductor, a semiconductor-on-insulator (SOI) substrate, or the like, which may be doped (e.g., with a p-type or an n-type dopant) or undoped. The substrate **50** may be a wafer, such as a silicon wafer. Generally, an SOI substrate is a layer of a semiconductor material formed on an insulator layer. The insulator layer may be, for example, a buried oxide (BOX) layer, a silicon oxide layer, or the like. The insulator layer is provided on a substrate, typically a silicon or glass substrate. Other substrates, such as a multi-layered or gradient substrate may also be used. In some embodiments, the semiconductor material of the substrate **50** may include silicon; germanium; a compound semiconductor including silicon carbide, gallium arsenide, gallium phosphide, indium phosphide, indium arsenide, and/or indium antimonide; an alloy semiconductor including silicon-germanium, gallium arsenide phosphide, aluminum indium arsenide, aluminum gallium arsenide, gallium indium arsenide, gallium indium phosphide, and/or gallium indium arsenide phosphide; or combinations thereof.

(13) The substrate **50** has a region **50N** and a region **50P**. The region **50N** can be for forming n-type devices, such as NMOS transistors, e.g., n-type FinFETs. The region **50P** can be for forming p-type devices, such as PMOS transistors, e.g., p-type FinFETs. The region **50N** may be physically separated from the region **50P** (as illustrated by divider **51**), and any number of device features (e.g., other active devices, doped regions, isolation structures, etc.) may be disposed between the region **50N** and the region **50P**.

(14) In FIG. **3**, fins **52** are formed in the substrate **50**. The fins **52** are semiconductor strips. In some embodiments, the fins **52** may be formed in the substrate **50** by etching trenches in the substrate **50**. The etching may be any acceptable etch process, such as a reactive ion etch (RIE), neutral beam etch (NBE), the like, or a combination thereof. The etch may be anisotropic.

(15) The fins may be patterned by any suitable method. For example, the fins may be patterned using one or more photolithography processes, including double-patterning or multi-patterning processes. Generally, double-patterning or multi-patterning processes combine photolithography and self-aligned processes, allowing patterns to be created that have, for example, pitches smaller than what is otherwise obtainable using a single, direct photolithography process. For example, in

one embodiment, a sacrificial layer is formed over a substrate and patterned using a photolithography process. Spacers are formed alongside the patterned sacrificial layer using a self-aligned process. The sacrificial layer is then removed, and the remaining spacers may then be used to pattern the fins. In some embodiments, the mask (or other layer) may remain on the fins **52**.

(16) In FIG. **4**, an insulation material **54** is formed over the substrate **50** and between neighboring fins **52**. The insulation material **54** may be an oxide, such as silicon oxide, a nitride, the like, or a combination thereof, and may be formed by a high density plasma chemical vapor deposition (HDP-CVD), a flowable CVD (FCVD) (e.g., a CVD-based material deposition in a remote plasma system and post curing to make it convert to another material, such as an oxide), the like, or a combination thereof. Other insulation materials formed by any acceptable process may be used. In the illustrated embodiment, the insulation material **54** is silicon oxide formed by a FCVD process. An anneal process may be performed once the insulation material is formed. In an embodiment, the insulation material **54** is formed such that excess insulation material **54** covers the fins **52**.

Although the insulation material **54** is illustrated as a single layer, some embodiments may utilize multiple layers. For example, in some embodiments a liner (not shown) may first be formed along a surface of the substrate **50** and the fins **52**. Thereafter, a fill material, such as those discussed above may be formed over the liner.

(17) In FIG. **5**, a removal process is applied to the insulation material **54** to remove excess insulation material **54** over the fins **52**. In some embodiments, a planarization process such as a chemical mechanical polish (CMP), an etch-back process, combinations thereof, or the like may be utilized. The planarization process exposes the fins **52** such that top surfaces of the fins **52** and the insulation material **54** are level after the planarization process is complete. In embodiments in which a mask remains on the fins **52**, the planarization process may expose the mask or remove the mask such that top surfaces of the mask or the fins **52**, respectively, and the insulation material **54** are level after the planarization process is complete.

(18) In FIG. **6**, the insulation material **54** is recessed to form Shallow Trench Isolation (STI) regions **56**. The insulation material **54** is recessed such that upper portions of fins **52** in the region **50N** and in the region **50P** protrude from between neighboring STI regions **56**. Further, the top surfaces of the STI regions **56** may have a flat surface as illustrated, a convex surface, a concave surface (such as dishing), or a combination thereof. The top surfaces of the STI regions **56** may be formed flat, convex, and/or concave by an appropriate etch. The STI regions **56** may be recessed using an acceptable etching process, such as one that is selective to the material of the insulation material **54** (e.g., etches the material of the insulation material **54** at a faster rate than the material of the fins **52**). For example, an oxide removal using, for example, dilute hydrofluoric acid (dHF) may be used.

(19) The process described with respect to FIGS. **2** through **6** is just one example of how the fins **52** may be formed. In some embodiments, the fins may be formed by an epitaxial growth process. For example, a dielectric layer can be formed over a top surface of the substrate **50**, and trenches can be etched through the dielectric layer to expose the underlying substrate **50**. Homoepitaxial structures can be epitaxially grown in the trenches, and the dielectric layer can be recessed such that the homoepitaxial structures protrude from the dielectric layer to form fins. Additionally, in some embodiments, heteroepitaxial structures can be used for the fins **52**. For example, the fins **52** in FIG. **5** can be recessed, and a material different from the fins **52** may be epitaxially grown over the recessed fins **52**. In such embodiments, the fins **52** comprise the recessed material as well as the epitaxially grown material disposed over the recessed material. In an even further embodiment, a dielectric layer can be formed over a top surface of the substrate **50**, and trenches can be etched through the dielectric layer. Heteroepitaxial structures can then be epitaxially grown in the trenches using a material different from the substrate **50**, and the dielectric layer can be recessed such that the heteroepitaxial structures protrude from the dielectric layer to form the fins **52**. In some embodiments where homoepitaxial or heteroepitaxial structures are epitaxially grown, the

epitaxially grown materials may be in situ doped during growth, which may obviate prior and subsequent implantations although in situ and implantation doping may be used together.

(20) Still further, it may be advantageous to epitaxially grow a material in region **50N** (e.g., an NMOS region) different from the material in region **50P** (e.g., a PMOS region). In various embodiments, upper portions of the fins **52** may be formed from silicon-germanium ($\text{Si.sub.xGe.sub.1-x}$, where x can be in the range of 0 to 1), silicon carbide, pure or substantially pure germanium, a III-V compound semiconductor, a II-VI compound semiconductor, or the like. For example, the available materials for forming III-V compound semiconductor include, but are not limited to, indium arsenide, aluminum arsenide, gallium arsenide, indium phosphide, gallium nitride, indium gallium arsenide, indium aluminum arsenide, gallium antimonide, aluminum antimonide, aluminum phosphide, gallium phosphide, and the like.

(21) Further in FIG. **6**, appropriate wells (not shown) may be formed in the fins **52** and/or the substrate **50**. In some embodiments, a P well may be formed in the region **50N**, and an N well may be formed in the region **50P**. In some embodiments, a P well or an N well are formed in both the region **50N** and the region **50P**.

(22) In the embodiments with different well types, the different implant steps for the region **50N** and the region **50P** may be achieved using a photoresist or other masks (not shown). For example, a photoresist may be formed over the fins **52** and the STI regions **56** in the region **50N**. The photoresist is patterned to expose the region **50P** of the substrate **50**, such as a PMOS region. The photoresist can be formed by using a spin-on technique and can be patterned using acceptable photolithography techniques. Once the photoresist is patterned, an n-type impurity implant is performed in the region **50P**, and the photoresist may act as a mask to substantially prevent n-type impurities from being implanted into the region **50N**, such as an NMOS region. The n-type impurities may be phosphorus, arsenic, antimony, or the like implanted in the region to a concentration of equal to or less than $10.\text{sup.18 cm.sup.-3}$, such as between about $10.\text{sup.16 cm.sup.-3}$ and about $10.\text{sup.18 cm.sup.-3}$. After the implant, the photoresist is removed, such as by an acceptable ashing process.

(23) Following the implanting of the region **50P**, a photoresist is formed over the fins **52** and the STI regions **56** in the region **50P**. The photoresist is patterned to expose the region **50N** of the substrate **50**, such as the NMOS region. The photoresist can be formed by using a spin-on technique and can be patterned using acceptable photolithography techniques. Once the photoresist is patterned, a p-type impurity implant may be performed in the region **50N**, and the photoresist may act as a mask to substantially prevent p-type impurities from being implanted into the region **50P**, such as the PMOS region. The p-type impurities may be boron, boron fluoride, indium, or the like implanted in the region to a concentration of equal to or less than $10.\text{sup.18 cm.sup.-3}$, such as between about $10.\text{sup.16 cm.sup.-3}$ and about $10.\text{sup.18 cm.sup.-3}$. After the implant, the photoresist may be removed, such as by an acceptable ashing process.

(24) After the implants of the region **50N** and the region **50P**, an anneal may be performed to repair implant damage and to activate the p-type and/or n-type impurities that were implanted. In some embodiments, the grown materials of epitaxial fins may be in situ doped during growth, which may obviate the implantations, although in situ and implantation doping may be used together.

(25) In FIG. **7**, a dummy dielectric layer **60** is formed on the fins **52**. The dummy dielectric layer **60** may be, for example, silicon oxide, silicon nitride, a combination thereof, or the like, and may be deposited or thermally grown according to acceptable techniques. A dummy gate layer **62** is formed over the dummy dielectric layer **60**, and a mask layer **64** is formed over the dummy gate layer **62**. The dummy gate layer **62** may be deposited over the dummy dielectric layer **60** and then planarized, such as by a CMP. The mask layer **64** may be deposited over the dummy gate layer **62**. The dummy gate layer **62** may be a conductive or non-conductive material and may be selected from a group including amorphous silicon, polycrystalline-silicon (polysilicon), polycrystalline silicon-germanium (poly-SiGe), metallic nitrides, metallic silicides, metallic oxides, and metals.

The dummy gate layer **62** may be deposited by physical vapor deposition (PVD), CVD, sputter deposition, or other techniques known and used in the art for depositing the selected material. The dummy gate layer **62** may be made of other materials that have a high etching selectivity from the etching of isolation regions. The mask layer **64** may include, for example, silicon nitride, silicon oxynitride, or the like. In this example, a single dummy gate layer **62** and a single mask layer **64** are formed across the region **50N** and the region **50P**. It is noted that the dummy dielectric layer **60** is shown covering only the fins **52** for illustrative purposes only. In some embodiments, the dummy dielectric layer **60** may be deposited such that the dummy dielectric layer **60** covers the STI regions **56**, extending between the dummy gate layer **62** and the STI regions **56**.

(26) FIGS. **8A** through **16B** illustrate various additional steps in the manufacturing of embodiment devices. FIGS. **8A** through **16B** illustrate features in either of the region **50N** and the region **50P**. For example, the structures illustrated in FIGS. **8A** through **16B** may be applicable to both the region **50N** and the region **50P**. Differences (if any) in the structures of the region **50N** and the region **50P** are described in the text accompanying each figure.

(27) In FIGS. **8A** and **8B**, the mask layer **64** (see FIG. **7**) may be patterned using acceptable photolithography and etching techniques to form masks **74**. The pattern of the masks **74** then may be transferred to the dummy gate layer **62**. In some embodiments (not illustrated), the pattern of the masks **74** may also be transferred to the dummy dielectric layer **60** by an acceptable etching technique to form dummy gates **72**. The dummy gates **72** cover respective channel regions **58** of the fins **52**. The pattern of the masks **74** may be used to physically separate each of the dummy gates **72** from adjacent dummy gates. The dummy gates **72** may also have a lengthwise direction substantially perpendicular to the lengthwise direction of respective epitaxial fins **52**.

(28) Further in FIGS. **8A** and **8B**, gate seal spacers **80** can be formed on exposed surfaces of the dummy gates **72**, the masks **74**, and/or the fins **52**. A thermal oxidation or a deposition followed by an anisotropic etch may form the gate seal spacers **80**. The gate seal spacers **80** may be formed of silicon oxide, silicon nitride, silicon oxynitride, or the like.

(29) After the formation of the gate seal spacers **80**, implants for lightly doped source/drain (LDD) regions (not explicitly illustrated) may be performed. In the embodiments with different device types, similar to the implants discussed above in FIG. **6**, a mask, such as a photoresist, may be formed over the region **50N**, while exposing the region **50P**, and appropriate type (e.g., p-type) impurities may be implanted into the exposed fins **52** in the region **50P**. The mask may then be removed. Subsequently, a mask, such as a photoresist, may be formed over the region **50P** while exposing the region **50N**, and appropriate type impurities (e.g., n-type) may be implanted into the exposed fins **52** in the region **50N**. The mask may then be removed. The n-type impurities may be the any of the n-type impurities previously discussed, and the p-type impurities may be the any of the p-type impurities previously discussed. The lightly doped source/drain regions may have a concentration of impurities of from about 10^{15} cm.⁻³ to about 10^{19} cm.⁻³. An anneal may be used to repair implant damage and to activate the implanted impurities.

(30) In FIGS. **9A** and **9B**, gate spacers **86** are formed on the gate seal spacers **80** along sidewalls of the dummy gates **72** and the masks **74**. The gate spacers **86** may be formed by conformally depositing an insulating material and subsequently anisotropically etching the insulating material. The insulating material of the gate spacers **86** may be silicon oxide, silicon nitride, silicon oxynitride, silicon carbonitride, a combination thereof, or the like. In some embodiments, the gate spacers **86** may have dangling bonds on outer surfaces.

(31) It is noted that the above disclosure generally describes a process of forming spacers and LDD regions. Other processes and sequences may be used. For example, fewer or additional spacers may be utilized, different sequence of steps may be utilized (e.g., the gate seal spacers **80** may not be etched prior to forming the gate spacers **86**, yielding “L-shaped” gate seal spacers, spacers may be formed and removed, and/or the like. Furthermore, the n-type and p-type devices may be formed using a different structures and steps. For example, LDD regions for n-type devices may be formed

prior to forming the gate seal spacers **80** while the LDD regions for p-type devices may be formed after forming the gate seal spacers **80**.

(32) In FIGS. **10A** and **10B** epitaxial source/drain regions **82** are formed in the fins **52** to exert stress in the respective channel regions **58**, thereby improving performance. The epitaxial source/drain regions **82** are formed in the fins **52** such that each dummy gate **72** is disposed between respective neighboring pairs of the epitaxial source/drain regions **82**. In some embodiments the epitaxial source/drain regions **82** may extend into, and may also penetrate through, the fins **52**. In some embodiments, the gate spacers **86** are used to separate the epitaxial source/drain regions **82** from the dummy gates **72** by an appropriate lateral distance so that the epitaxial source/drain regions **82** do not short out subsequently formed gates of the resulting FinFETs.

(33) The epitaxial source/drain regions **82** in the region **50N**, e.g., the NMOS region, may be formed by masking the region **50P**, e.g., the PMOS region, and etching source/drain regions of the fins **52** in the region **50N** to form recesses in the fins **52**. Then, the epitaxial source/drain regions **82** in the region **50N** are epitaxially grown in the recesses. The epitaxial source/drain regions **82** may include any acceptable material, such as appropriate for n-type FinFETs. The epitaxial source/drain regions **82** in the region **50N** may have surfaces raised from respective surfaces of the fins **52** and may have facets.

(34) The epitaxial source/drain regions **82** can be formed using an epitaxy process such as a cyclic deposition-etch (CDE) process. The CDE process includes a number of repeated cycles, such as in a range from 2 cycles to w cycles. Each cycle of the CDE process includes a deposition process followed by an etch process. In some embodiments, the deposition process of the CDE process includes a chemical vapor deposition (CVD) process, such as reduced pressure chemical vapor deposition (RPCVD), low pressure CVD (LPCVD), the like, or a combination thereof. In some embodiments, the process is RPCVD.

(35) For example, if the fin **52** is silicon, the epitaxial source/drain regions **82** in the region **50N** may include materials exerting a tensile strain in the channel region **58**, such as silicon phosphide (SiP), silicon phosphorous carbide (SiPC), or the like. The epitaxial source/drain regions **82** are in situ doped with the conductivity dopant species (e.g., an n-type dopant, like phosphorous in embodiments described herein). A silicon source precursor gas can be used for the RPCVD. The silicon source precursor gas can be a silicon-rich precursor gas, such as including silane (SiH_4), dichlorosilane (SiH_2Cl_2 , DCS), trichlorosilane (SiHCl_3), disilane (Si_2H_6), a combination thereof, or the like. A flow rate of the silicon source precursor gas of the RPCVD can be in a range from about 40 sccm to about 1000 sccm.

(36) The RPCVD process can also include a phosphorous source precursor gas. The phosphorous source precursor gas can include phosphine (PH_3), phosphorus oxychloride, another phosphorous-containing precursor, and/or any combination thereof. In some embodiments, a ratio of the silicon source precursor to the phosphorous source precursor gas is in a range from about 50 to about 300. Carrier gases, such as hydrogen (H_2), can be mixed with the precursors in either of the above embodiments. In some embodiments, the RPCVD process uses a silicon-rich precursor gas, such as dichlorosilane (DCS), and a phosphorous source precursor gas, such as phosphine, with a hydrogen carrier gas. In some embodiments, a ratio of the silicon source precursor to the carrier gas is in a range from about 2:1 to about 10:1.

(37) In some embodiments, the use of dichlorosilane (DCS) as the silicon source precursor and hydrogen as the carrier gas during the epitaxial growth of epitaxial source/drain regions **82** may produce rounded top profiles of the epitaxial source/drain regions **82**. DCS may allow for even growth of crystalline silicon in most or all lattice planes and hydrogen may inhibit the growth of crystalline silicon in the 100 plane (horizontal growth) because the hydrogen attaches to dangling bonds in the 100 plane while not inhibiting growth in the 110 and 111 lattice planes. Using the combination of DCS as the silicon source precursor and hydrogen as the carrier gas may allow for

controlled growth of rounded top profiles. An amount of hydrogen in a range of about 2 L to about 10 L may be used in order to produce the rounded top profiles.

(38) In some embodiments, the use of hydrogen as a carrier gas may prevent or reduce the formation of nodule defects, such as on the gate spacers **86** which may comprise a nitride such as, e.g. SiN. Nodule defects may be roughly spherical growths that form during the epitaxial growth process due to the precursor gases attaching disproportionately to dangling bonds on, such as e.g. the gate spacers **86**. Nodule defects can lower device performance by altering the shape of the epitaxial source/drain regions **82**. The hydrogen carrier gas may passivate the surface of the gate spacers **86** by terminating on the surface of the gate spacers **86**, which may prevent nodule defects from forming during the growth of the epitaxial source/drain regions **82**.

(39) A pressure of the RPCVD can be equal to or less than about 300 Torr, such as in a range from about 50 Torr to about 300 Torr. In some cases, a pressure of smaller than 50 Torr for the RPCVD may provide an insufficient dopant concentration. In some cases, a pressure of greater than 300 Torr for the RPCVD may lead to selective loss. The pressure can vary depending on the particular process being used. A temperature of the RPCVD can be in a range from about 650° C. to about 750° C. In some embodiments, the parameters may vary based on the process.

(40) The epitaxial source/drain regions **82** can have various concentrations of the conductivity dopant species. When phosphorous is implemented as the conductivity dopant species, e.g., from epitaxial growth with phosphorous, a concentration of phosphorous in the epitaxial source/drain regions **82** can be in a range from about 1×10^{21} atoms/cm³ to about 4×10^{21} atoms/cm³.

(41) In some embodiments, the deposition process forms an epitaxial layer, such as SiP, in the recesses formed in the fins **52** and an amorphous material on non-crystalline surfaces. In some embodiments, after the deposition process, a post-deposition purge operation is used to remove the deposition gases from the process chamber. An inert gas, such as He, Ar, or Ne, may be used in this operation to purge the deposition gases from the process chamber. Once the deposition gases are removed from the chamber, the etch process follows.

(42) The etching (or partial etching) process of the CDE process removes the amorphous material and may also remove a portion of the deposited epitaxial layer. The remaining epitaxial layer forms the epitaxial source/drain regions **82**. The etch process can be an in situ etch process performed in the chamber of the deposition process. In some embodiments, an etch gas is flowed into the chamber to etch the amorphous material. Etch gases such as chlorine (Cl₂), hydrochloric acid (HCl), hydrofluoric acid (HF), hydrobromic acid (HBr), or the like can be used. A pressure during the etch process can be equal to or less than about 300 Torr, such as in a range from about 5 Torr to about 300 Torr. A temperature during the etch process can be in range from about 625° C. to about 750° C. In some examples, after the etching process, a purge operation follows to remove the etching gases from the chamber.

(43) The etching process can remove the amorphous material at a greater rate than the epitaxial material. This may be done with an etch selective to the amorphous material such as, e.g. an HCl etch, performed at a temperature in a range of about 625° C. to about 750° C. and at a pressure in a range of about 5 Torr to about 300 Torr. Therefore, the epitaxial material remains on, e.g., the surfaces of the epitaxial source/drain regions **82** after the deposition-etch cycle. The deposition-etch cycle may be repeated a number of times until a desired thickness of the epitaxial source/drain regions **82** is reached. By removing appropriate portions of amorphous material, desired shapes of the epitaxial source/drain regions **82** may be produced. As a result, such repeated deposition-etch process is called a cyclic deposition-etch (CDE) process.

(44) The epitaxial source/drain regions **82** in the region **50P**, e.g., the PMOS region, may be formed by masking the region **50N**, e.g., the NMOS region, and etching source/drain regions of the fins **52** in the region **50P** to form recesses in the fins **52**. Then, the epitaxial source/drain regions **82** in the region **50P** are epitaxially grown in the recesses. The epitaxial source/drain regions **82** may include

any acceptable material, such as appropriate for p-type FinFETs. For example, if the fin **52** is silicon, the epitaxial source/drain regions **82** in the region **50P** may comprise materials exerting a compressive strain in the channel region **58**, such as silicon-germanium, boron doped silicon-germanium, germanium, germanium tin, or the like. The epitaxial source/drain regions **82** in the region **50P** may also have surfaces raised from respective surfaces of the fins **52** and may have facets.

(45) As a result of the epitaxy processes used to form the epitaxial source/drain regions **82** in the region **50N** and the region **50P**, upper surfaces of the epitaxial source/drain regions **82** have facets which expand laterally outward beyond sidewalls of the fins **52**. In some embodiments, these facets cause adjacent source/drain regions **82** of a same FinFET to merge as illustrated by FIG. **10C**. In other embodiments, adjacent source/drain regions **82** remain separated after the epitaxy process is completed as illustrated by FIG. **10D**. In the embodiments illustrated in FIGS. **10C** and **10D**, gate spacers **86** are formed covering a portion of the sidewalls of the fins **52** that extend above the STI regions **56** thereby blocking the epitaxial growth. In some other embodiments, the spacer etch used to form the gate spacers **86** may be adjusted to remove the spacer material to allow the epitaxially grown region to extend to the surface of the STI region **56**.

(46) In FIG. **10C**, the upper facets of adjacent source/drain regions **82** have merged, producing a merged epitaxial source/drain region **82** with a first portion disposed in a first fin **52**, and a second portion disposed in a second fin **52**. Top surfaces of the first and second fins **52** are shown in outline in FIG. **10C**. The first portion and the second portion of the merged epitaxial source/drain region **82** are joined at a merging boundary **104**. The merging boundary **104** may extend from a lowest point of a valley between the fins **52** in the top surface of the merged epitaxial source/drain region **82** to a vertex of the bottom surface of the merged epitaxial source/drain region **82** located between the fins **55**.

(47) The merged epitaxial source/drain region **82** may have a first width $W_{sub.1}$ in a range of about 40 nm to about 70 nm. It is advantageous for the first width $W_{sub.1}$ to be in the range of about 40 nm to about 70 nm for improving both process yields and device properties. The first width $W_{sub.1}$ being less than about 40 nm may lead to lower yield and device loss. The first width $W_{sub.1}$ being greater than about 70 nm may lead to lower yield and device loss.

(48) The merged epitaxial source/drain region **82** comprises a first subregion **82A** extending from a highest point of the merged epitaxial source/drain region **82** to a highest point of the merging boundary **104**. The top surface of the first subregion **82A** comprises two rounded top profiles. In some embodiments, the use of dichlorosilane (DCS) as the silicon source precursor and hydrogen as the carrier gas during the epitaxial growth of the epitaxial source/drain regions **82** may produce rounded top profiles of the merged epitaxial source/drain regions **82**. DCS may allow for even growth of crystalline silicon in most or all lattice planes and hydrogen may inhibit the growth of crystalline silicon in the 100 plane (horizontal growth) because the hydrogen attaches to dangling bonds in the 100 plane while not inhibiting growth in the 110 and $1\bar{1}0$ lattice planes. Using the combination of DCS as the silicon source precursor and hydrogen as the carrier gas may allow for controlled growth of rounded top profiles. An amount of hydrogen in a range of about 2 L to about 10 L may be used in order to produce the rounded top profiles.

(49) The rounded top profiles are located above opposite sides of the merging boundary **104**. The first subregion **82A** may have a first height $H_{sub.1}$ in a range of less than about 10 nm. In some embodiments, the top surface of the first subregion **82A** has a valley between the fins **52**. Because the merging boundary **104** is located above the widest diameter of the merged epitaxial source/drain region **82**, the valley between the rounded top profiles of the merged epitaxial source/drain region **82** may be relatively shallow, in comparison with a valley formed with a merging boundary located in a lower position. The valley between the rounded top profiles of the merged epitaxial source/drain region **82** may provide a more stable landing area for source/drain contacts. Because the valley between the rounded top profiles of the merged epitaxial source/drain

region **82** is relatively shallow, formation of source/drain contacts such as, e.g., metal plugs is less likely to break through the merged facets of the epitaxial source/drain region **82** between the fins **52**.

(50) The valley may have a height equal to the first height $H_{sub.1}$ from its lowest point to its highest point. It is advantageous for the first height $H_{sub.1}$ to be in the range of about less than 10 nm for improving both process yields and device properties. The first height $H_{sub.1}$ being greater than about 10 nm may lead to lower yield and device loss.

(51) The merged epitaxial source/drain region **82** further comprises a second subregion **82B** extending from the highest point of the merging boundary **104** to a lowest point of the merging boundary **104**. The second subregion **82B** may have a second height $H_{sub.2}$ in a range of about 18 nm to about 28 nm. It is advantageous for the second height $H_{sub.2}$ to be in the range of about 18 nm to about 28 nm for improving both process yields and device properties. The second height $H_{sub.2}$ being less than about 18 nm may lead to lower yield and device loss. The second height $H_{sub.2}$ being greater than about 28 nm may lead to lower yield and device loss. A ratio of the first height $H_{sub.1}$ to the second height $H_{sub.2}$ may be in a range of about 0:28 to about 10:18. It is advantageous for the ratio $H_{sub.1}:H_{sub.2}$ to be in the range of about 0:28 to about 10:18 for improving both process yields and device properties. The ratio $H_{sub.1}:H_{sub.2}$ being greater than about 5:9 may lead to lower yield and device loss.

(52) The merged epitaxial source/drain region **82** further comprises a third subregion **82C** extending from the lowest point of the merging boundary **104** to the top surface of the STI regions **56**. The third subregion **82C** may have a third height $H_{sub.3}$ in a range of about 25 nm to about 40 nm. It is advantageous for the third height $H_{sub.3}$ to be in the range of about 25 nm to about 40 nm for improving both process yields and device properties. The third height $H_{sub.3}$ being less than about 25 nm may lead to device loss. A ratio of the first height $H_{sub.1}$ to the third height $H_{sub.3}$ may be in a range of about 0:25 to about 10:40. It is advantageous for the ratio $H_{sub.1}:H_{sub.3}$ to be in the range of about 0:28 to about 10:40 for improving both process yields and device properties.

(53) Embodiments of the merged epitaxial source/drain regions **82** may provide advantages. The merged epitaxial source/drain regions **82** may be useful for high current (HC) SRAM transistors. HC SRAM transistors comprising multiple fin FinFETs with merged epitaxial source/drain regions may be useful for high speed circuits to achieve good performance in speed. The rounded top profiles of the epitaxial source/drain regions **82** may prevent failures to merge by increasing the horizontal widths of the neighboring epitaxial source/drain regions **82** near their top surfaces. The rounded top profiles of the merged epitaxial source/drain region **82** may form a better landing for source/drain contact plugs, as described below with respect to FIG. **16C**. The higher intra-fin merge height of the merged epitaxial source/drain regions **82** may improve device performance by reducing the capacitance effect of the source/drain regions.

(54) In FIG. **10D**, the upper facets of adjacent source/drain regions **82** have not merged, producing adjacent non-merged epitaxial source/drain region **82** disposed in neighboring fins **52**. The top surfaces of each non-merged epitaxial source/drain region **82** comprise respective rounded top profiles. The rounded top profiles of the non-merged epitaxial source/drain region **82** may be formed by using hydrogen as the carrier gas. The non-merged epitaxial source/drain regions **82** may have a second width $W_{sub.2}$ measured between opposite sidewalls of the epitaxial source/drain regions **82** at their distance of greatest separation in a range of about 25 nm to about 40 nm. It is advantageous for the second width $W_{sub.2}$ to be in the range of about 25 nm to about 40 nm for improving both process yields and device properties. The second width $W_{sub.2}$ being less than about 25 nm may lead to lower yield and device loss. The second width $W_{sub.2}$ being greater than about 40 nm may lead to lower yield and device loss.

(55) In accordance with some embodiments as illustrated by FIG. **10E**, the epitaxial source/drain regions **82** may be formed with rounded bottom profiles rather than the faceted bottom profiles as

illustrated by FIG. 10B. The shapes of the bottom profiles of the epitaxial source/drain regions **82** may be adjusted by, e.g., controlling the shapes of the recesses etched into the top surfaces of the fins **52**.

(56) The non-merged epitaxial source/drain regions **82** may have a third width $W_{sub.3}$ measured 10 nm below a highest point of the non-merged epitaxial source/drain regions **82**. The third width $W_{sub.3}$ may be in a range of about 15 nm to about 35 nm. It is advantageous for the third width $W_{sub.3}$ to be in the range of about 15 nm to about 35 nm for improving both process yields and device properties. The third width $W_{sub.3}$ being less than about 15 nm may lead to lower yield and device loss. The third width $W_{sub.3}$ being greater than about 35 nm may lead to lower yield and device loss.

(57) A ratio of the second width $W_{sub.2}$ to the third width $W_{sub.3}$ may be in a range of about 25:15 to about 40:35. It is advantageous for the ratio $W_{sub.2}:W_{sub.3}$ to be in the range of about 25:15 to about 40:35 for improving both process yields and device properties. The ratio $W_{sub.2}:W_{sub.3}$ being less than about 5:3 may lead to lower yield and device loss.

(58) The non-merged epitaxial source/drain regions **82** may have a fourth height $H_{sub.4}$ measured from a bottom point of the non-merged epitaxial source/drain regions **82** to a top point of the non-merged epitaxial source/drain regions **82** in a range of about 40 nm to about 60 nm. It is advantageous for the fourth height $H_{sub.4}$ to be in the range of about 40 nm to about 60 nm for improving both process yields and device properties. The fourth height $H_{sub.4}$ being less than about 40 nm may lead to lower yield and device loss. The fourth height $H_{sub.4}$ being greater than about 60 nm may lead to lower yield and device loss.

(59) Embodiments of the non-merged epitaxial source/drain regions **82** may provide advantages. The non-merged epitaxial source/drain regions **82** may be useful for high density (HD) SRAM transistors. HD SRAM transistors comprising single fin FinFETs with non-merged epitaxial source/drain regions between adjacent FinFETs may be useful for reducing power usage and area. This may achieve minimal cell standby leakage current as well a reduction in cell size, which is advantageous for miniaturization. HD SRAM cells may have a smaller pitch than the pitch of HC SRAM cells. In embodiments in which the non-merged epitaxial source/drain regions **82** are used in high density (HD) SRAM devices, the rounded top shapes of the epitaxial source/drain regions **82** may improve fin coverage of HD SRAM structures and prevent merging, or shorts, between neighboring HD SRAM source/drain regions by creating slimmer epitaxial source/drain shapes. The slim epitaxial source/drain region shapes of the HD SRAM structures may improve device performance by reducing the capacitance effect of the source/drain regions.

(60) As illustrated in FIGS. 10C and 10D, in some embodiments the epitaxial source/drain regions **82** may extend below a surface level with the top surface of the STI regions **56**. In other embodiments, the epitaxial source/drain regions **82** may be level with the top surface of the STI regions **56**. In still other embodiments, bottom surfaces of the epitaxial source/drain regions **82** may be located above the surface level with the top surface of the STI regions **56**.

(61) In some embodiments, the rounded top profiles of the epitaxial source/drain regions **82** may reduce highly doped SiP source/drain region loss for downstream MEOL and BEOL processes. Production throughput may improve by about 20% due to a higher epitaxial growth rate and a reduced transition time. Enlarged highly doped source/drain volume may lead to reduced source/drain contact plug resistance.

(62) In FIGS. 11A, 11B, 11C, and 11D, a first interlayer dielectric (ILD) **88** is deposited over the structures illustrated in FIGS. 10A, 10B, 10C, and 10D. The first ILD **88** may be formed of a dielectric material, and may be deposited by any suitable method, such as CVD, plasma-enhanced CVD (PECVD), or FCVD. Dielectric materials may include phospho-silicate glass (PSG), boro-silicate glass (BSG), boron-doped phospho-silicate glass (BPSG), undoped silicate glass (USG), or the like. Other insulation materials formed by any acceptable process may be used. In some embodiments, a contact etch stop layer (CESL) **87** is disposed between the first ILD **88** and the

epitaxial source/drain regions **82**, the masks **74**, and the gate spacers **86**. The CESL **87** may comprise a dielectric material, such as, silicon nitride, silicon oxide, silicon oxynitride, or the like, having a different etch rate than the material of the overlying first ILD **88**.

(63) In FIGS. **12A**, **12B**, **12C**, and **12D**, a planarization process, such as a CMP, may be performed to level the top surface of the first ILD **88** with the top surfaces of the dummy gates **72** or the masks **74**. The planarization process may also remove the masks **74** on the dummy gates **72**, and portions of the gate seal spacers **80** and the gate spacers **86** along sidewalls of the masks **74**. After the planarization process, top surfaces of the dummy gates **72**, the gate seal spacers **80**, the gate spacers **86**, and the first ILD **88** are level. Accordingly, the top surfaces of the dummy gates **72** are exposed through the first ILD **88**. In some embodiments, the masks **74** may remain, in which case the planarization process levels the top surface of the first ILD **88** with the top surfaces of the masks **74**.

(64) In FIGS. **13A**, **13B**, **13C**, and **13D**, the dummy gates **72**, and the masks **74** if present, are removed in an etching step(s), so that recesses **90** are formed. Portions of the dummy dielectric layer **60** in the recesses **90** may also be removed. In some embodiments, only the dummy gates **72** are removed and the dummy dielectric layer **60** remains and is exposed by the recesses **90**. In some embodiments, the dummy dielectric layer **60** is removed from recesses **90** in a first region of a die (e.g., a core logic region) and remains in recesses **90** in a second region of the die (e.g., an input/output region). In some embodiments, the dummy gates **72** are removed by an anisotropic dry etch process. For example, the etching process may include a dry etch process using reaction gas(es) that selectively etch the dummy gates **72** without etching the first ILD **88** or the gate spacers **86**. Each recess **90** exposes and/or overlies a channel region **58** of a respective fin **52**. Each channel region **58** is disposed between neighboring pairs of the epitaxial source/drain regions **82**. During the removal, the dummy dielectric layer **60** may be used as an etch stop layer when the dummy gates **72** are etched. The dummy dielectric layer **60** may then be optionally removed after the removal of the dummy gates **72**.

(65) In FIGS. **14A**, **14B**, **14C**, and **14D**, gate dielectric layers **92** and gate electrodes **94** are formed for replacement gates. FIG. **14E** illustrates a detailed view of region **89** of FIG. **14B**. Gate dielectric layers **92** are deposited conformally in the recesses **90**, such as on the top surfaces and the sidewalls of the fins **52** and on sidewalls of the gate seal spacers **80**/gate spacers **86**. The gate dielectric layers **92** may also be formed on the top surface of the first ILD **88**. In accordance with some embodiments, the gate dielectric layers **92** comprise silicon oxide, silicon nitride, or multilayers thereof. In some embodiments, the gate dielectric layers **92** include a high-k dielectric material, and in these embodiments, the gate dielectric layers **92** may have a k value greater than about 7.0, and may include a metal oxide or a silicate of hafnium, aluminum, zirconium, lanthanum, manganese, barium, titanium, lead, and combinations thereof. The formation methods of the gate dielectric layers **92** may include Molecular-Beam Deposition (MBD), ALD, PECVD, and the like. In embodiments where portions of the dummy gate dielectric **60** remains in the recesses **90**, the gate dielectric layers **92** include a material of the dummy gate dielectric **60** (e.g., SiO₂).

(66) The gate electrodes **94** are deposited over the gate dielectric layers **92**, respectively, and fill the remaining portions of the recesses **90**. The gate electrodes **94** may include a metal-containing material such as titanium nitride, titanium oxide, tantalum nitride, tantalum carbide, cobalt, ruthenium, aluminum, tungsten, combinations thereof, or multi-layers thereof. For example, although a single layer gate electrode **94** is illustrated in FIG. **14B**, the gate electrode **94** may comprise any number of liner layers **94A**, any number of work function tuning layers **94B**, and a fill material **94C** as illustrated by FIG. **14E**. After the filling of the recesses **90**, a planarization process, such as a CMP, may be performed to remove the excess portions of the gate dielectric layers **92** and the material of the gate electrodes **94**, which excess portions are over the top surface of the ILD **88**. The remaining portions of material of the gate electrodes **94** and the gate dielectric layers **92** thus form replacement gates of the resulting FinFETs. The gate electrodes **94** and the gate

dielectric layers **92** may be collectively referred to as a “gate stack.” The gate and the gate stacks may extend along sidewalls of a channel region **58** of the fins **52**.

(67) The formation of the gate dielectric layers **92** in the region **50N** and the region **50P** may occur simultaneously such that the gate dielectric layers **92** in each region are formed from the same materials, and the formation of the gate electrodes **94** may occur simultaneously such that the gate electrodes **94** in each region are formed from the same materials. In some embodiments, the gate dielectric layers **92** in each region may be formed by distinct processes, such that the gate dielectric layers **92** may be different materials, and/or the gate electrodes **94** in each region may be formed by distinct processes, such that the gate electrodes **94** may be different materials. Various masking steps may be used to mask and expose appropriate regions when using distinct processes.

(68) In FIGS. **15A**, **15B**, **15C**, and **15D**, a second ILD **108** is deposited over the first ILD **88**. In some embodiment, the second ILD **108** is a flowable film formed by a flowable CVD method. In some embodiments, the second ILD **108** is formed of a dielectric material such as PSG, BSG, BPSG, USG, or the like, and may be deposited by any suitable method, such as CVD and PECVD. In accordance with some embodiments, before the formation of the second ILD **108**, the gate stack (including a gate dielectric layer **92** and a corresponding overlying gate electrode **94**) is recessed, so that a recess is formed directly over the gate stack and between opposing portions of gate spacers **86**, as illustrated in FIGS. **15A** and **15B**. A gate mask **96** comprising one or more layers of dielectric material, such as silicon nitride, silicon oxynitride, or the like, is filled in the recess, followed by a planarization process to remove excess portions of the dielectric material extending over the first ILD **88**. The subsequently formed gate contacts **110** (FIGS. **16A** and **16B**) penetrate through the gate mask **96** to contact the top surface of the recessed gate electrode **94**.

(69) In FIGS. **16A**, **16B**, **16C**, and **16D**, gate contacts no and source/drain contacts **112** are formed through the second ILD **108** and the first ILD **88** in accordance with some embodiments. Openings for the source/drain contacts **112** are formed through the first and second ILDs **88** and **108**, and openings for the gate contact no are formed through the second ILD **108** and the gate mask **96**. The openings may be formed using acceptable photolithography and etching techniques. A liner, such as a diffusion barrier layer, an adhesion layer, or the like, and a conductive material are formed in the openings. The liner may include titanium, titanium nitride, tantalum, tantalum nitride, or the like. The conductive material may be copper, a copper alloy, silver, gold, tungsten, cobalt, aluminum, nickel, or the like. A planarization process, such as a CMP, may be performed to remove excess material from a surface of the ILD **108**. The remaining liner and conductive material form the source/drain contacts **112** and gate contacts no in the openings. An anneal process may be performed to form a silicide at the interface between the epitaxial source/drain regions **82** and the source/drain contacts **112**. The source/drain contacts **112** are physically and electrically coupled to the epitaxial source/drain regions **82**, and the gate contacts no are physically and electrically coupled to the gate electrodes **94**. The source/drain contacts **112** and gate contacts no may be formed in different processes, or may be formed in the same process. Although shown as being formed in the same cross-sections, it should be appreciated that each of the source/drain contacts **112** and gate contacts no may be formed in different cross-sections, which may avoid shorting of the contacts. The rounded top profiles of the epitaxial source/drain region **82** may provide a better, relatively flat landing area for source/drain contacts **112** for both high current (HC) and high density (HD) SRAM structures, as shown in FIGS. **16C** and **16D**. This may further reduce resistance of the source/drain contacts **112**. FIG. **16E** illustrates embodiments following from FIG. **10E** with epitaxial source/drain regions **82** having rounded bottom profiles.

(70) In some embodiments, the shapes of the epitaxial source/drain region **82** may result in improved characteristics. Nodule defects may be reduced by choice of an optimized gas ratio and carrier gas. The rounded top shapes of the epitaxial source/drain regions may prevent non-merging of intra-fin epitaxial regions, e.g. silicon phosphide (SiP), of neighboring source/drain regions used in HC SRAM devices. The rounded top shapes of the epitaxial source/drain regions may improve

fin coverage of HD SRAM and prevent merging, or shorts, between neighboring HD SRAM source/drain regions by creating slimmer epitaxial source/drain shapes. Larger landing areas for contacts for both HC and HD SRAM structures may be provided by the rounded top profiles of the epitaxial regions, which may further reduce contact resistance. Enlarged highly doped source/drain volume may lead to reduced source/drain contact plug resistance. The rounded top profiles may reduce highly doped SiP source/drain region loss for downstream MEOL and BEOL processes. The higher intra-fin merge height of the HC SRAM structures and the slim epitaxial source/drain region shapes may improve device performance by reducing the capacitance effect.

(71) In accordance with an embodiment, a semiconductor device includes: a first fin and a second fin, the first fin and the second fin extending from a substrate; a shallow trench isolation (STI) region disposed on the substrate between the first and second fins; and an epitaxial source/drain region, wherein the epitaxial source/drain region includes a first portion grown on the first fin and a second portion grown on the second fin, and wherein the first portion and the second portion are joined at a merging boundary. The epitaxial source/drain region further includes: a first subregion extending from a location level with a highest point of the epitaxial source/drain region to a location level with a highest point of the merging boundary, wherein the first subregion has a first height, and wherein a top surface of the first subregion includes a valley between the first fin and the second fin; a second subregion extending from the location level with the highest point of the merging boundary to a location level with a lowest point of the merging boundary, wherein the second subregion has a second height, and wherein the first height is less than the second height; and a third subregion extending from the location level with the lowest point of the merging boundary to a location level with a top surface of the STI region, wherein the third subregion has a third height, and wherein the first height is less than the third height and the second height is less than the third height. In an embodiment, the epitaxial source/drain region includes SiP. In an embodiment, a concentration of phosphorus in the epitaxial source/drain region is in a range from 1×10^{21} atoms/cm³ to 4×10^{21} atoms/cm³. In an embodiment, the epitaxial source/drain region extends below the location level with the top surface of the STI region. In an embodiment, the epitaxial source/drain region has a width in a range of 40 nm to 70 nm. In an embodiment, the first height is less than 10 nm. In an embodiment, the device is part of a memory array. In an embodiment, a ratio of the first height to the second height is less than 5:9. In an embodiment, a ratio of the first height to the third height is less than 1:4.

(72) In accordance with another embodiment, a semiconductor device includes: a high density (HD) circuit area on a substrate including an HD memory cell and a high current (HC) circuit area on a substrate including an HC memory cell with an operating speed greater than the HD memory cell, a pitch of fins in the HD memory cell being smaller than the HC memory cell, the HD memory cell including: a first fin extending from the substrate; a first epitaxial source/drain region on the first fin; a second fin extending from the substrate and adjacent the first fin; a third fin extending from the substrate and adjacent the first fin; a second epitaxial source/drain region on the second fin; and a third epitaxial source/drain region on the third fin, wherein the third epitaxial source/drain region is physically separated from the second epitaxial source/drain region and the first epitaxial source/drain region; and a high current (HC) circuit area on a substrate including an HC memory cell with an operating speed greater than the HD memory cell, a pitch of fins in the HD memory cell is smaller than the HC memory cell, the HC memory cell including: a fourth fin and a fifth fin extending from the substrate; and a fourth epitaxial source/drain region, wherein the fourth epitaxial source/drain region includes a first portion grown on the fourth fin, wherein the fourth epitaxial source/drain region includes a second portion grown on the fifth fin, wherein the first portion and the second portion are joined at a merging boundary, and wherein the fourth epitaxial source/drain region further includes: a first subregion extending from a location level with a highest point of the fourth epitaxial source/drain region to a location level with a highest point of the merging boundary; a second subregion extending from the location level with the highest point

of the merging boundary to a location level with a lowest point of the merging boundary; and a third subregion extending from the location level with the lowest point of the merging boundary to a location level with a lowest point of the fourth epitaxial source/drain region. In an embodiment, the first epitaxial source/drain region has a first height measured from a location level with a bottom point of the first epitaxial source/drain region to a location level with a highest point of the first epitaxial source/drain region, wherein the first epitaxial source/drain region has a first width measured between points of opposite sidewalls of the first epitaxial source/drain region at their greatest separation, and wherein a ratio of the first width to the first height is in a range of 5:12 to 1:1. In an embodiment, a first width of the first epitaxial source/drain region is in a range of 25 nm to 40 nm. In an embodiment, a first height of the first epitaxial source/drain region is in a range of 40 nm to 60 nm. In an embodiment, the first epitaxial source/drain region has a second width measured 10 nm below a highest point of the first epitaxial source/drain region, wherein the second width is in a range of 15 nm to 35 nm.

(73) In accordance with yet another embodiment, a method for forming a semiconductor device includes: forming shallow trench isolation (STI) regions adjacent a first fin and a second fin disposed on a substrate; forming a first recess in the first fin; forming a second recess in a second fin; and epitaxially growing a first source/drain region in the first recess and a second source/drain region in the second recess, wherein the first and second source/drain regions include respective rounded top profiles, wherein the first source/drain region and the second source/drain region merge at a merging boundary to form a merged source/drain region, wherein the merged source/drain region has a first height measured from a location level with a highest point of the merged source/drain region to a location level with a highest point of the merging boundary, wherein the merged source/drain region has a second height measured from the location level with the highest point of the merging boundary to a location level with a lowest point of the merging boundary, wherein the merged source/drain region has a third height measured from the location level with the lowest point of the merging boundary to a location level with the top surface of the STI regions, wherein the first height is less than the second height, wherein the first height is less than the third height, and wherein the second height is less than the third height. In an embodiment, epitaxially growing the source/drain region includes a reduced pressure chemical vapor deposition (RPCVD) process. In an embodiment, the RPCVD process is performed at a temperature in a range from 650° C. to 750° C. In an embodiment, epitaxially growing the source/drain regions includes a cyclic-deposition etch (CDE) process. In an embodiment, epitaxially growing the source/drain regions includes using a silicon-rich precursor gas, a phosphorus source precursor gas, and a carrier gas, the carrier gas including hydrogen. In an embodiment, a ratio of the silicon-rich precursor gas to the carrier gas is in a range from 2:1 to 10:1.

(74) In accordance with yet another embodiment, a method for forming a semiconductor device includes: forming an isolation region on a substrate, a first semiconductor protrusion extending from the substrate on a first side of the isolation region, a second semiconductor protrusion extending from the substrate on a second side of the isolation region, the second side being opposite the first side; forming a first recess in the first semiconductor protrusion and forming a second recess in the second semiconductor protrusion; and epitaxially growing a first source/drain region in the first recess and a second source/drain region in the second recess, wherein the first source/drain region and the second source/drain region merge at a merging boundary, the first source/drain region having a first rounded top profile in a cross-sectional view, the second source/drain region having a second rounded top profile in the cross-sectional view, a valley between the first rounded top profile and the second rounded top profile having a depth less than 10 nm. In an embodiment, forming the first rounded top profile and the second rounded top profile includes flowing dichlorosilane and hydrogen. In an embodiment, the hydrogen attaches to dangling bonds in respective two planes of the first source/drain region and the second source/drain region. In an embodiment, 2 L to 10 L of hydrogen are used while flowing the hydrogen. In an

embodiment, the method further includes etching trenches in the substrate, wherein etching the trenches forms the first semiconductor protrusion and the second semiconductor protrusion. In an embodiment, the method further includes: forming a gate structure over the first semiconductor protrusion and the second semiconductor protrusion; forming gate spacers adjacent the gate structure; and while epitaxially growing the first source/drain region and the second source/drain region, flowing hydrogen as a carrier gas, wherein the hydrogen passivates the surface of the gate spacers. In an embodiment, epitaxially growing the first source/drain region and the second source/drain region is performed with a reduced pressure chemical vapor deposition (RPCVD), the RPCVD being performed at a pressure in a range of 50 Torr to 300 Torr. In an embodiment, the first source/drain region and the second source/drain region include silicon phosphide. In an embodiment, a concentration of phosphorus in the first source/drain region and the second source/drain region is in a range from 1×10^{21} atoms/cm³ to 4×10^{21} atoms/cm³.
-3.

(75) In accordance with yet another embodiment, a method for forming a semiconductor device includes: forming a first recess in a first semiconductor protrusion, forming a second recess in a second semiconductor protrusion, and forming a third recess in a third semiconductor protrusion, the first semiconductor protrusion, the second semiconductor protrusion, and the third semiconductor protrusion extending from a substrate in a high density (HD) circuit area of the substrate, the second semiconductor protrusion being adjacent a first side of the first semiconductor protrusion, the third semiconductor protrusion being adjacent a second side of the first semiconductor protrusion, the second side being opposite the first side; forming a fourth recess in a fourth semiconductor protrusion and forming a fifth recess in a fifth semiconductor protrusion, the fourth semiconductor protrusion and the fifth semiconductor protrusion extending from the substrate in a high current (HC) circuit area of the substrate, the fifth semiconductor protrusion being adjacent the fourth semiconductor protrusion, wherein a first pitch of semiconductor protrusions in the HD circuit area is smaller than a second pitch of semiconductor protrusions in the HC circuit area; epitaxially growing a first source/drain region in the first recess, a second source/drain region in the second recess, and a third source/drain region in the third recess, wherein the first source/drain region remains separated from the second source/drain region and from the third source/drain region, wherein the first source/drain region, the second source/drain region, and the third source/drain region have respective rounded top profiles in a cross-sectional view; epitaxially growing a fourth source/drain region in the fourth recess and a fifth source/drain region in the fifth recess, wherein the fourth source/drain region and the fifth source/drain region merge at a merging boundary), wherein the fourth source/drain region and the fifth source/drain region have respective rounded top profiles in the cross-sectional view; forming an HD memory cell in the HD circuit area, wherein the HD memory cell includes the first semiconductor protrusion, the second semiconductor protrusion, the third semiconductor protrusion, the first source/drain region, the second source/drain region, and the third source/drain region; and forming an HC memory cell in the HC circuit area, wherein the HC memory cell includes the fourth semiconductor protrusion, the fifth semiconductor protrusion, and the fourth source/drain region. In an embodiment, an operating speed of the HC memory cell is greater than an operating speed of the HD memory cell. In an embodiment, the first source/drain region has a first height measured from a location level with a bottom point of the first source/drain region to a location level with a highest point of the first source/drain region, wherein the first source/drain region has a first width measured between points of opposite sidewalls of the first source/drain region at their greatest separation, and wherein a ratio of the first width to the first height is in a range of 5:12 to 1:1. In an embodiment, the method further includes etching trenches in the substrate, wherein etching the trenches forms the first semiconductor protrusion, the second semiconductor protrusion, the third semiconductor protrusion, the fourth semiconductor protrusion, and the fifth semiconductor protrusion.

(76) The foregoing outlines features of several embodiments so that those skilled in the art may

better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A method for forming a semiconductor device, the method comprising: forming a first fin; forming a second fin adjacent to the first fin; epitaxially growing a first source/drain region in a first recess of the first fin; and epitaxially growing a second source/drain region in a second recess of the second fin, wherein the first source/drain region has a first rounded top profile and the second source/drain region has a second rounded top profile, wherein forming the first rounded top profile and forming the second rounded top profile comprises flowing dichlorosilane and hydrogen, wherein the hydrogen attaches to dangling bonds in a first **100** plane of the first source/drain region and a second **100** plane of the second source/drain region, wherein the first source/drain region and the second source/drain region merge at a merging boundary to form a merged source/drain region.
2. The method of claim 1, wherein the epitaxially growing the first source/drain region comprises a reduced pressure chemical vapor deposition (RPCVD) process.
3. The method of claim 2, wherein the RPCVD process is performed at a temperature in a range from 650° C. to 750° C.
4. The method of claim 1, wherein the epitaxially growing the second source/drain region comprises a cyclic-deposition etch process.
5. The method of claim 1, further comprising doping the first source/drain region and the second source/drain region with conductivity dopant species.
6. The method of claim 1, wherein a ratio of dichlorosilane gas to hydrogen is in a range from 2:1 to 10:1.
7. The method of claim 5, wherein the epitaxially growing the first source/drain region with the first rounded top profile and the epitaxially growing the second source/drain region with the second rounded top profile further comprises the flowing of the hydrogen to an amount in a range of 2 L to 10 L.
8. A method for forming a semiconductor device, the method comprising: forming a gate structure over a first semiconductor protrusion extending from a substrate on a first side of an isolation region and over a second semiconductor protrusion extending from the substrate on a second side of the isolation region opposite the first side; forming gate spacers adjacent the gate structure; epitaxially growing a first source/drain region in a first recess in the first semiconductor protrusion, wherein the first source/drain region has a first rounded top profile; and epitaxially growing a second source/drain region in a second recess in the second semiconductor protrusion, wherein the first source/drain region and the second source/drain region merge at a merging boundary, wherein during the epitaxially growing the first source/drain region and during the epitaxially growing the second source/drain region, flowing hydrogen, wherein the hydrogen passivates a surface of the gate spacers.
9. The method of claim 8, wherein the second source/drain region has a second rounded top profile.
10. The method of claim 8, wherein the hydrogen attaches to dangling bonds in a 100 plane of the first source/drain region.
11. The method of claim 8, wherein 2 L to 10 L of hydrogen are utilized during the flowing the hydrogen.
12. The method of claim 8, further comprising etching trenches in the substrate, wherein etching

the trenches forms the first semiconductor protrusion and the second semiconductor protrusion.

13. The method of claim 8, wherein the epitaxially growing the first source/drain region comprises using a phosphorus source precursor gas.

14. The method of claim 8, wherein the epitaxially growing the first source/drain region is performed with a reduced pressure chemical vapor deposition (RPCVD).

15. The method of claim 14, wherein the RPCVD is performed at a temperature in a range from 650° C. to 750° C.

16. The method of claim 8, wherein the second source/drain region comprise silicon phosphide.

17. A method for forming a semiconductor device, the method comprising: forming a first fin; forming a second fin adjacent to the first fin; and epitaxially growing a first source/drain region in a first recess of the first fin and epitaxially growing a second source/drain region in a second recess of the second fin, wherein the step of epitaxially growing the first source/drain region and the second source/drain region comprises flowing dichlorosilane and hydrogen to produce a first rounded top profile for the first source/drain region and a second rounded top profile for the second source/drain region, wherein the first source/drain region and the second source/drain region merge at a merging boundary, wherein flowing dichlorosilane inhibits epitaxial growth of the first source/drain region and the second source/drain region in a horizontal direction.

18. The method of claim 17, further comprising forming gate spacers over the first fin and wherein flowing hydrogen passivates a surface of the gate spacers.

19. The method of claim 17, wherein the dichlorosilane and hydrogen are flowed at a ratio in a range from 2:1 to 10:1.
